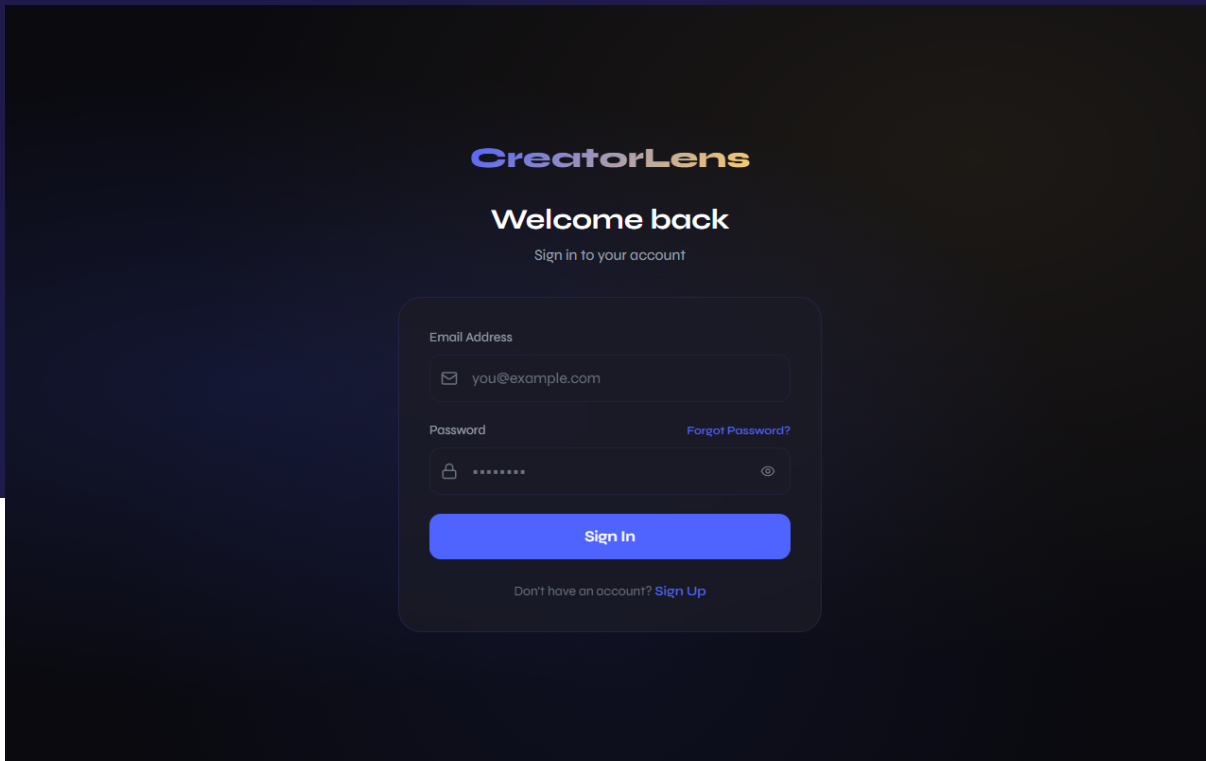


CollabBridge

AI-Powered Brand & Creator Collaboration Platform



System Workflow & Integration Guide

A comprehensive, step-by-step handbook covering authentication, brand campaign setups, creator profiles, AI scoring and auditing metrics.

PLATFORM DOCUMENTATION V1.0

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1. Platform Introduction

CollabBridge is a state-of-the-art marketing collaboration marketplace connecting brands with influential creators. The platform resolves standard friction points in influencer marketing by embedding verification, real-time messaging, dispute-free milestone tracking, and advanced AI-driven profile analytics.

Brands use the platform to post detailed campaign requests (budgets, deliverables, rules), and find matching creators. Creators join to establish a digital media kit, discover paid work, apply to active campaigns, and collaborate securely. Administrators supervise the ecosystem, vetting creators and monitoring campaigns for compliance.

Technology Stack Details

- **Frontend Application:** Next.js 14, React Redux Toolkit, Tailwind CSS, Recharts for analytics UI.
- **Backend Services:** Node.js (Express), Node-Mailer for alerts, Socket.io for persistent chat channels.
- **Database Engine:** MySQL managed via Sequelize ORM with fully normalized relationship models.
- **AI Core Systems:** Custom JavaScript algorithm analyzing engagement, consistency, and location metrics.

2. Core System Architecture

CollabBridge is structured around five primary relational entities: Users, Campaigns, Applications, Messages, and Payments. A relational MySQL schema ensures referential integrity across registrations, applications, and payments.

Key Core Database Entities

- **Users:** Stores credentials, profile info, bio, social links, role (brand/creator/admin) and AI ratings.
- **Campaigns:** Holds brand campaign definitions, budget, deliverables, platform requirements, and status.
- **Applications:** Tracks creator applications to campaigns, including pitches, bids, and states (pending/accepted).
- **Messages:** Stores persistent chat history between brands and creators, complete with socket indicators.
- **Payments:** Manages contract terms, total deposit, milestone values, and releases via escrow.

Core REST API Router References

Method	API Endpoint	Purpose & Action Scope
POST	/api/auth/register	Register creator, brand, or admin profile
POST	/api/auth/login	Authenticate profile, return JWT authorization token
GET	/api/users/creators	Query list of creators with platform & rating filters
POST	/api/users/analyze/:id	Trigger background AI profile validation score
POST	/api/campaigns	Create new brand campaign proposal
POST	/api/applications	Submit creator campaign application & bid
GET	/api/messages/:id	Retrieve historical chat logs for chat session
PUT	/api/admin/users/:id/verify	Mark creator profile verified by admin audit

3. Brand Workflow: Finding Creators

A brand's journey begins with creator discovery. CollabBridge provides an advanced search engine allowing brands to find and collaborate with verified creators. The interface includes a sidebar for navigation, a search bar, filters for niches, followers, and engagement, and a grid of creator profiles with detailed metrics and social links.

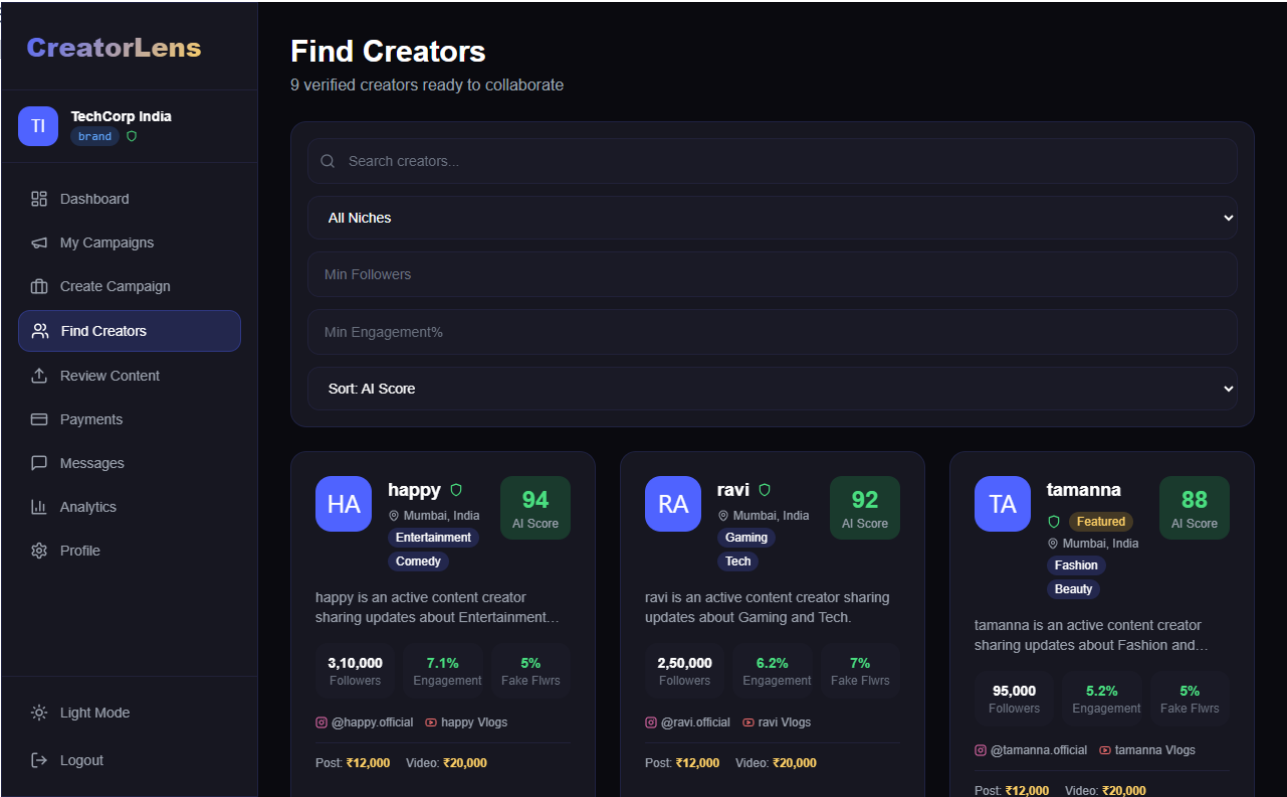


Figure 1. Creator discovery search portal showing filtering options and rating cards.

Creator Search Features & Metrics:

- **Granular Niche Filters:** Instantly narrow down search results to specific niches (e.g., Tech or Beauty).
- **Aggregated Scoring:** Review aggregated ratings based on past campaigns, rating stars, and completed contracts.
- **Verification Badges:** Differentiate between standard user registrations and verified platform creators.

4. Brand Workflow: AI Profile Analysis

To protect brands from fraudulent accounts and inflated metrics, CollabBridge includes a built-in AI Analysis System. The AI Analysis System is designed to detect and flag suspicious activity, ensuring that brands are only connected to authentic and high-quality creators.

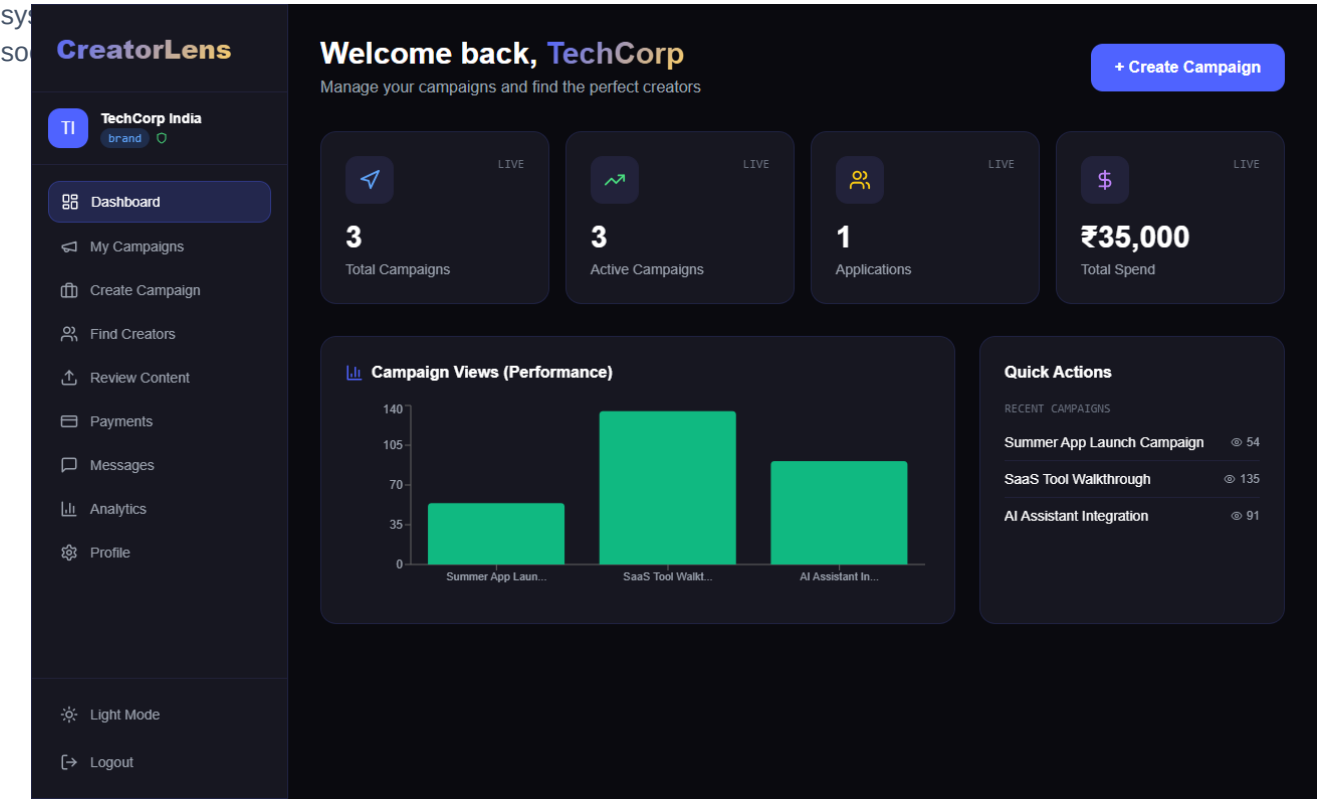


Figure 2. Brand dashboard UI detailing campaign progress, total budget spent, and key metrics.

AI Analysis Audit Metrics Explained:

- **Engagement Rate:** Evaluates historical likes/comments against follower counts to detect active audience ratio.
- **Audience Authenticity:** Uses profile verification heuristics to estimate the percentage of real vs fake followers.
- **Posting Consistency:** Measures active posting schedules to determine if creator keeps up an active presence.
- **Audience Demographics:** Extracts geographical distribution details to match brand campaign target regions.

5. Brand Workflow: Campaign Posting

Posting a campaign allows brands to invite applications passively. The creator dashboard lists active campaign briefs and allows brands to create new campaigns.

CreatorLens

Ti

TechCorp India

brand

Dashboard

My Campaigns

Create Campaign

Find Creators

Review Content

Payments

Messages

Analytics

Profile

Light Mode

Logout

Create Campaign

Launch a new brand collaboration campaign

Campaign Details

Campaign Title *

Summer Collection Launch 2024

Description *

Describe your campaign, product, and what you're looking for in a creator...

Target & Platforms

Content Niche

Tech Fashion Travel Food Fitness Beauty Gaming Education Finance

Lifestyle

Platforms

Instagram Youtube Twitter Tiktok All

Min Followers Required

Min Engagement Rate (%)

Figure 3. Campaign setup form interface detailing budgets, channels, and briefs.

Key Campaign Parameters:

- **Deliverables Brief:** Detailed text instructions specifying what the creator needs to produce (videos, posts).
- **Budget Allocation:** Brands specify a total budget, which will be held in platform escrow during collaboration.
- **Platform Selection:** Specify which social channels (e.g. YouTube, Instagram) the promotion must run on.

6. Creator Workflow: Applying & Earning

Creators log in to a tailored view highlighting active campaigns matching their category. From their dashboard, creators can track their progress and earnings.

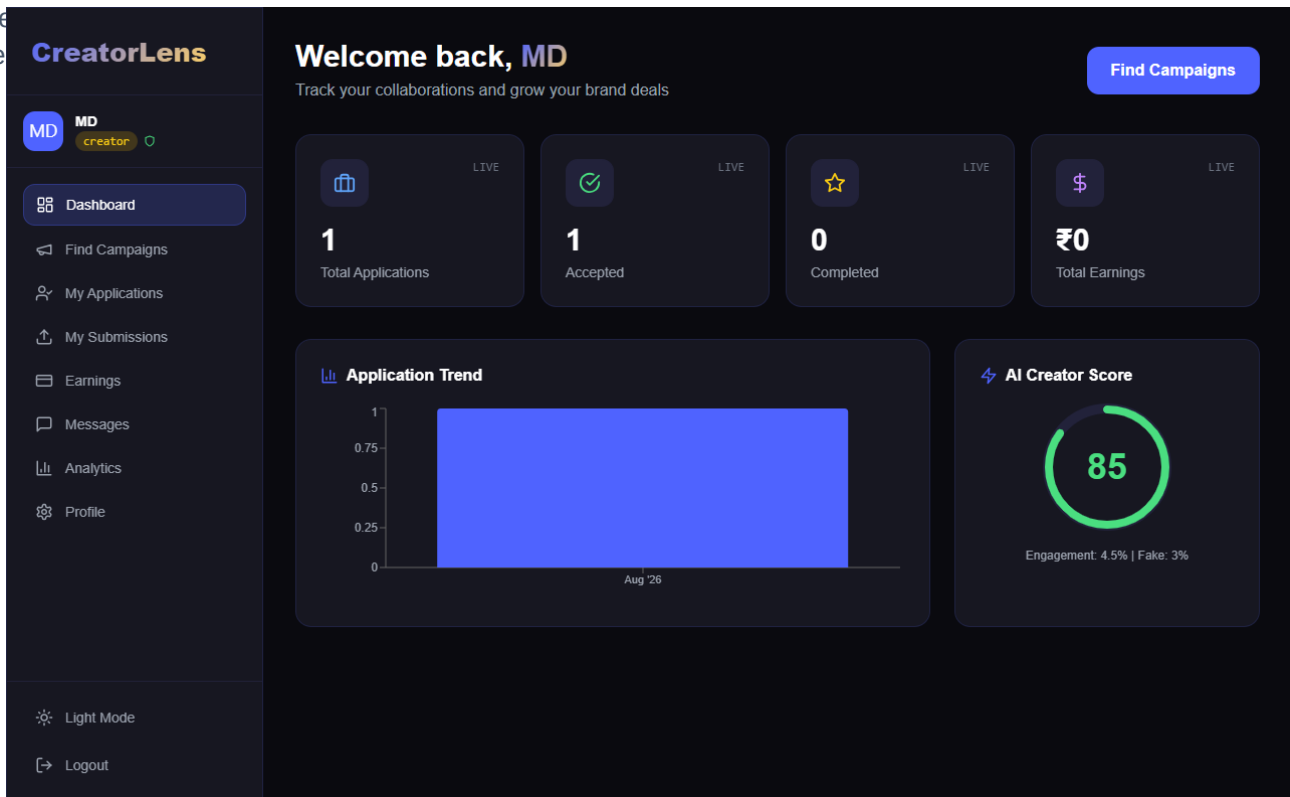


Figure 4. Creator dashboard interface displaying monthly earnings, active milestones, and applications.

Creator Application Lifecycle:

- **Explore Briefs:** Browse list of active campaigns, searching by platform tags or keyword filters.
- **Submit Pitch:** Write a proposal letter outlining creative ideas, adding links to past portfolio pieces.
- **Milestone Trackers:** Upon application acceptance, view active contract phases, submissions, and payments.

7. Messaging & Administrative Control

CollabBridge integrates a real-time messaging system built on Socket.io. This enables prompt negotiations, asset

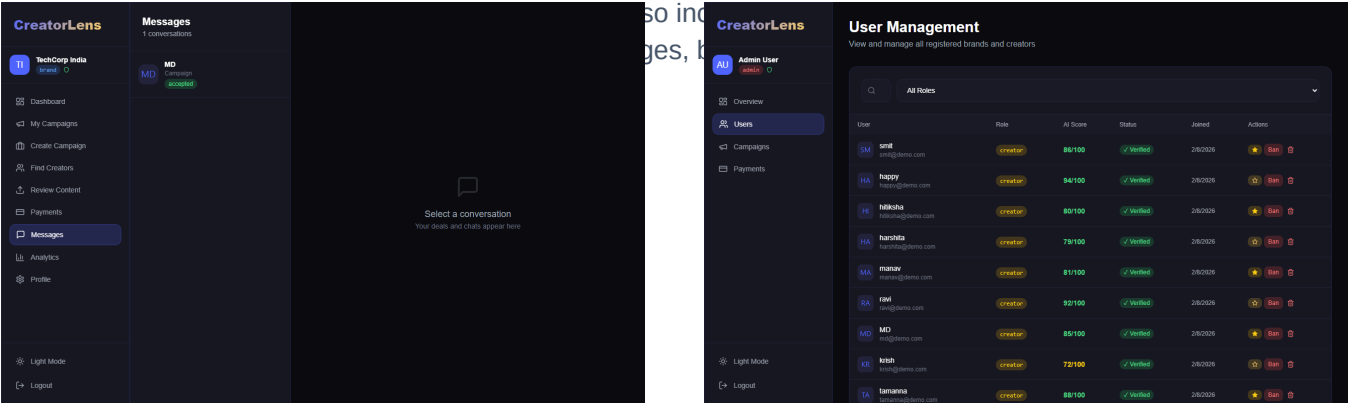


Figure 5. Left: Chat interface for direct messages. Right: Platform Admin Oversight dashboard.

Collaboration and Management Channels:

- **Socket.io Instant Chats:** Real-time, low-latency text channels for creators and brands to negotiate terms.
- **User Moderation Engine:** Admin console enabling single-click creator profile verification and user banning.
- **Platform Statistics Panel:** High-level metrics tracking total system revenue, registrations, and chats.

IMPORTANT INTEGRATION RUNTIME NOTE

Before running the admin verification or initiating chat sockets, ensure that node seed.js has successfully completed execution to provision demo credentials in the local MySQL instance.